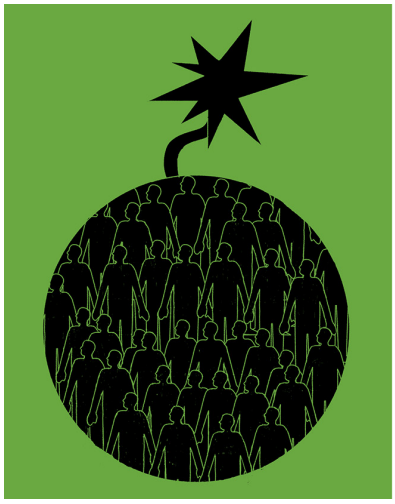


POPULATION DYNAMICS BECOME CHAOTIC WHEN THE RATE OF REPRODUCTION SOARS

CHAOTIC POPULATION CHANGE

IN CONTEXT

KEY FIGURE

Robert May (1936–2020)

BEFORE

1798 Thomas Malthus argues that human populations will increase at an ever-faster rate, inevitably causing suffering.

1845 Belgian demographist Pierre-François Verhulst argues that checks to population growth will increase in line with population growth itself.

AFTER

1987 Per Bak, Chao Tang, and Kurt Wiesenfeld, a research team in New York, describe “self-organized criticality”—elements within a system interacting spontaneously to produce change.

2014 Japanese ecologist George Sugihari uses a chaos theory approach called empirical dynamic modeling to produce a more accurate estimate of salmon numbers in Canada’s Fraser River.

Chaos theory—the idea that predictions are limited by time and the nonlinear nature of behavior—took hold in the 1960s. American meteorologist Edward Lorenz observed the effect in weather patterns, and